

Chi-Square Tests of Independence

For the tests of independence of two categorical variables, the chi-square test statistic is

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

Where *Observed* are the actual counts, and *Expected counts* are calculated as

So, the test takes several steps:

$$\text{expected count} = \frac{\text{row total} \times \text{column total}}{n}$$

1. Arrange the observed counts in a table

Observed counts

Stratum	Sampled	NotAllowed	Total
Small	57	6	63
Medium	17	5	22
Large	5	1	6
Total	79	12	91

2. Follow the formula for expected counts. For example, this command
=D3*B\$6/\$D\$6

gives the expected count in cell B3 (sampled small) = 54.69.

All my row totals are in column D, and all the row totals in row 3, so I added dollars \$ and dragged the formula around the whole table.

Expected counts

Stratum	Sampled	NotAllowed	Total
Small	54.69	8.31	63
Medium	19.10	2.90	22
Large	5.21	0.79	6
Total	79	12	91

3. Follow the formula for the chi-square components. For example,



$$=(B3-G3)^2/G3$$

computes the component for the cell “sampled small” = 0.10.

Chi-square components

Stratum	Sampled	NotAllowed
Small	0.10	0.64
Medium	0.23	1.52
Large	0.01	0.06

4. The chi-square statistic is the sum of chi-square components,

$$=SUM(B10:C12)$$

5. The P-value can be computed as

$$=CHISQ.DIST.RT(E9,2)$$

The option RT gives the needed right-tail probability, and 2 stays for the degrees of freedom = (r-1)(c-1). You can also use the command

$$=1-CHISQ.DIST(E9,2,1)$$

	A	B	C	D	E	F	G	H	I	J
1	Observed counts					Expected counts				
2	Stratum	Sampled	NotAllowed	Total		Stratum	Sampled	NotAllowe	Total	
3	Small	57	6	63		Small	54.69	8.31	63	
4	Medium	17	5	22		Medium	19.10	2.90	22	
5	Large	5	1	6		Large	5.21	0.79	6	
6	Total	79	12	91		Total	79	12	91	
7										
8	Chi-square components					Chi-square statistic				
9	Stratum	Sampled	NotAllowed		2.55					
10	Small	0.10	0.64		P-value					
11	Medium	0.23	1.52		0.27928					
12	Large	0.01	0.06		Conclusion: insufficient evidence of any dependence					

Steps 3-4-5 can also be accomplished by just one command

$$=CHITEST(B3:C5,G3:H5)$$

It returns the P-value of our chi-square test.

	A	B	C	D	E	F	G	H	I
1	Observed counts					Expected counts			
2	Stratum	Sampled	NotAllowed	Total		Stratum	Sampled	NotAllowe	Total
3	Small	57	6	63		Small	54.69	8.31	63
4	Medium	17	5	22		Medium	19.10	2.90	22
5	Large	5	1	6		Large	5.21	0.79	6
6	Total	79	12	91		Total	79	12	91
7									
8	=CHITEST(B3:C5,G3:H5)			0.27928					

P-value

